

GAWorld: Building Persistent and Situated LLM Agent Societies

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2026

Abstract

LLM agents are increasingly used to study social interaction, yet many demonstrations remain short-lived, weakly situated, or difficult to audit as scientific instruments. We present GAWorld, research infrastructure for building persistent and situated LLM agent societies. GAWorld connects profile-grounded agents, structured state, file-backed memory, planning, action, and reflection to a shared urban environment, social relations, economic and work processes, information exposure, and controlled interventions. Its execution model combines a configurable cognition pipeline with event and plugin boundaries while preserving compatibility with directly integrated mechanisms during an incremental architectural migration. The system exposes command-line, web, interview, replay, distributed communication, and experiment-comparison interfaces, together with heterogeneous artifacts that permit inspection from aggregate summaries back to agent and subsystem traces. We audit both the implementation and the existing result archive, using memory, policy, information, and economy–wellbeing cases to demonstrate research affordances and failure visibility. These artifacts establish system capabilities, not population realism, causal policy effects, or prediction of real cities. We therefore frame GAWorld as a modular instrument for constructing and auditing artificial society experiments, and make explicit the validation, reproducibility, and ethical work required before stronger social claims are warranted.

1 Introduction

Large language models (LLMs) make it possible to construct agents that interpret natural-language situations, maintain plans, and generate socially legible action. This has renewed interest in generative agents and in using LLMs to simulate individual or collective behavior (Park et al., 2023, 2024; Argyle et al., 2023). A convincing conversation, however, is not yet an artificial society. Society-level research requires agents to persist across simulated time, act from locations and relationships in a shared world, encounter consequences of earlier actions, and leave evidence from which a researcher can reconstruct what happened. It also requires mechanisms outside the language model—time, space, exposure, institutions, resource constraints, and interventions—to remain explicit enough to inspect and vary.

These requirements create a systems problem. Persistence cannot reside only in a model context window: profiles, state, episodic experience, relationships, and schedules must survive across ticks and runs. Situatedness cannot be reduced to scenery in a prompt: movement, co-location, local conditions, and social ties must constrain later computation. Experimental control requires scenario configuration and intervention points that are distinct from the agent’s generated narrative. Finally, scientific use requires provenance below a polished transcript or aggregate plot. A researcher should be able to ask which mechanism produced a state change, whether every treatment completed, and which model and configuration generated an archived trace.

GAWorld addresses this problem as research infrastructure for building persistent and situated LLM agent societies. It connects a continuing agent lifecycle—profile, perception, memory and recall, planning and action, reflection, state update, and storage—to an urban and societal substrate. That substrate includes a city graph and local physical conditions, social relationships and life events, economic state, work, information exposure, and scheduled interventions. A simulation clock and event boundaries connect individual cognition to day-level and society-level evolution. The resulting unit of study is not an isolated model response but a configured run whose local decisions, shared events, persistent state, and derived comparisons can be inspected together.

The system is also an account of incremental research-software modularization. GAWorld began with tightly connected domain paths and is being migrated toward a small execution kernel, configurable cognition stages, an event bus, and lifecycle-managed plugins. The current implementation contains working examples of these

boundaries while retaining direct calls, shared dictionaries, subsystem-owned files, and hook-based compatibility paths. We document that mixed state rather than presenting an aspirational architecture as complete. This distinction is important for reuse and for science: a mechanism that is configurable is not necessarily isolated, and an extension that writes a metric is not necessarily on the causal path of an agent action.

This paper makes four contributions:

1. **A persistent and situated agent lifecycle.** We describe how GAWorld joins relatively stable profiles, mutable structured state, file-backed memory, intentions, habits, relationships, interests, and skills to a configurable per-tick cognition pipeline and a shared simulation clock. The design makes continuity and context inspectable without treating them as evidence of psychological fidelity.
2. **A composable urban-society substrate.** We show how city, environment, social-network, economic, work, policy, and information mechanisms contribute to a common run. Explicit coupling and provenance support experiments that cross individual and aggregate levels while making clear where calibration and mechanism validation remain necessary.
3. **An extensible execution and research architecture.** We present the configured stage pipeline, simulation context, event bus, plugin registry, controller and recorder skeletons, compatibility paths, and the command-line, dashboard, interview, replay, and relay interfaces built around them. A migration-state audit separates wired services, partial integrations, and planned boundaries.
4. **An inspectable experimentation workflow with bounded evidence.** We connect scenario configuration, matched-run comparison, structured artifacts, and GAWorld-Bench to an evidence ledger. Existing memory, policy, information, and economy–wellbeing archives demonstrate what the system can record and expose; incomplete and diagnostic artifacts are reported alongside them rather than converted into claims of social validity.

GAWorld is not presented as a digital twin, a representative synthetic population, or a forecasting system. Its current artifacts do not establish that generated agents reproduce particular people, that aggregate trajectories match a real city, or that a simulated intervention estimates a real policy effect. We instead treat the system as a scientific instrument whose outputs must be evaluated at multiple levels and whose reproducibility is a prerequisite for substantive interpretation. This framing places software architecture, experimental design, and validity boundaries in the same account.

The remainder of the paper proceeds from this distinction. We first state the design goals and position GAWorld relative to generative-agent, language-agent, social-evaluation, and agent-based-modeling research. We then describe the agent lifecycle, execution architecture, memory and cognition, and societal substrate. Subsequent sections cover experiment and deployment interfaces, audit the existing capability cases, and state the validation, ethical, and development boundaries that govern intended use.

2 Design Goals

GAWorld is intended as a research instrument rather than a claim that a language model is, by itself, a valid population model. Its design goals therefore concern properties that a researcher can inspect in the running system. Each property below has a concrete mechanism in the present implementation and a boundary that prevents the mechanism from being read as an empirical guarantee.

2.1 Persistence across simulated time

An agent should carry consequences from one decision context into later ones, rather than being reconstructed from a stateless prompt at every tick. GAWorld combines a mutable state record with file-backed episodic memory, daily logs and diaries, schedules, and longer-lived relationship and growth records. A stateful run restores the day counter and per-agent memories, while the memory lifecycle can retrieve, consolidate, summarize, and decay stored episodes. This makes persistence testable: a researcher can inspect both the state before and after a tick and the files used in later recall. Persistence does not establish psychological fidelity. Several memories and summaries are language-model outputs, configuration can disable lifecycle operations, and the archived cross-phase evaluation discussed later in the paper is incomplete.

2.2 Situatedness in a shared world

Actions should be conditioned on where and when an agent acts and on what is locally observable. The simulation clock exposes day, time, and tick index; the city map represents named locations and connections; and environment,

local-physical, social, policy, and information mechanisms contribute context to the cognition loop. Movement updates location before later stages use the new setting, and social influence is computed over the current network rather than over an unconstrained cast of characters. This is situated computation, not a calibrated digital twin: map detail, physical dynamics, and agent access to information are abstractions selected by configuration.

2.3 Composability without concealing control flow

Researchers should be able to change an experimental mechanism without rewriting the complete agent loop. GAWorld provides three bounded extension surfaces: a configurable stage sequence for cognition, an event bus with observe/contribute/filter semantics, and a registry for lifecycle-managed plugins. The audited built-in set contains nine plugins: interventions, skills, interests, life events, economy, local physical perception, real work, dynamic behavior, and spatial preferences. Composability is not unrestricted hot swapping: stages exchange a shared dictionary, while memory, environment, and social-network logic retain direct compatibility paths.

2.4 Controllability for counterfactual experiments

A research run should expose explicit points at which a scenario or intervention is introduced. Configuration fixes profiles, city assets, random seed, mechanism flags, schedules, and policy events; the intervention plugin contributes a recommendation feed through the event bus and records its domain-specific metrics. Baseline/event workflows can then align traces produced under different scenario specifications. The kernel also defines a Controller API for validators and named runtime interventions. Structured movement requests pass through its validator chain, and the standard runtime interventions update agent state, update live configuration, or queue agent removal with audit records. Other selected actions are not universally mediated, so configuration, scheduled events, hooks, and experiment scripts remain important control surfaces.

2.5 Inspectability from decision to artifact

A claim about an artificial society should be traceable to artifacts below the level of a prose summary. GAWorld writes per-agent logs and memories, environment timelines, state histories, plugin-owned outputs, and experiment comparisons; its interfaces expose portions of those records for inspection and replay. The kernel Recorder additionally implements a clock-stamped JSONL stream and records Controller interventions and denials. That unified stream is not yet the main action/trace path: the established simulator continues to write several specialized formats directly. Inspection therefore requires an artifact ledger that names the producing path and schema, and replayability must not be inferred merely from the presence of logs.

2.6 Backward-compatible modularization

The system must remain usable while a monolithic research prototype is decomposed. A typed `Agent` view shares, rather than copies, the legacy dictionary; `EventBus` preserves the legacy hook signature; and the 12-stage pipeline was carved from the existing per-agent step while retaining the shared step dictionary. These adapters allow migrated and legacy modules to coexist and make migration state observable. They also retain weakly typed surfaces and shared mutable state. Backward compatibility is thus a transition strategy, not evidence that modularization is complete or that every old configuration produces an identical trajectory.

3 Related Work

GAWorld draws on four bodies of work that answer different parts of the artificial-society problem. Its contribution is a system integration and an audited set of research surfaces; we do not claim that the underlying ideas of memory, tool-using agents, social evaluation, or agent-based validation are unique to GAWorld.

3.1 Generative agents and simulated individuals

Generative Agents demonstrated that an LLM-driven architecture combining an experience stream, retrieval, reflection, and planning can sustain socially legible behavior in a sandbox community (Park et al., 2023). That work provides a central precedent for persistent social characters and for studying emergent interaction through an inspectable environment. Later work grounds agents in extensive self-report data and evaluates how well responses

predict held-out answers from the same individuals (Park et al., 2024). Research on “silicon samples” similarly explores conditioning language models to approximate distributions of survey responses (Argyle et al., 2023). These studies motivate profile grounding and longitudinal state, but they also distinguish two validation targets: internally coherent simulated inhabitants and externally evaluated models of particular people or populations.

GAWorld is oriented toward constructing the surrounding research system. It adds configurable urban, social, economic, work, information, and policy mechanisms; multiple operational interfaces; and explicit experiment and artifact workflows around a persistent-agent loop. Its archived profiles and traces have not undergone the held-out, person-level validation used in work on self-report-grounded agents. Accordingly, GAWorld adopts persistence as a system property and treats fidelity as an open empirical claim.

3.2 Language-agent cognition and architecture

LLM-agent research has developed reusable patterns for coupling language generation to action. ReAct interleaves reasoning and environment-facing actions (Yao et al., 2023); Reflexion stores verbal feedback to improve later attempts (Shinn et al., 2023); and Voyager combines iterative prompting, environment feedback, and an accumulating skill library in an open-ended embodied setting (Wang et al., 2024). MemoryBank studies long-term memory with retrieval and forgetting mechanisms (Zhong et al., 2024), while CoALA offers a conceptual organization of language agents in terms of memory, action space, and decision procedures (Sumers et al., 2024). CAMEL, in turn, examines role-conditioned communication among language agents (Li et al., 2023).

These systems and abstractions inform GAWorld’s separation of structured state, episodic storage, retrieval, planning, action, reflection, and learned skills. GAWorld differs in scope rather than by proposing a new generic reasoning algorithm: it connects such components to a persistent multi-agent society and to a clocked, file-backed experimental runtime. Its twelve-stage pipeline, event bus, and plugin registry expose replacement and observation points, but the present migration remains mixed with direct and hook-based paths. Thus GAWorld contributes an implemented integration for social simulation, not a claim of a universally superior cognitive architecture.

3.3 Social-agent tasks and evaluation

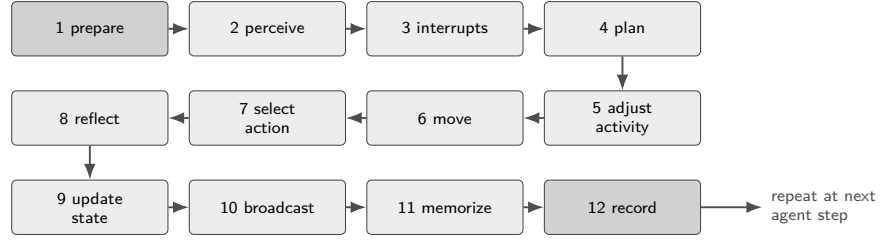
AgentBench evaluates LLM agents across interactive environments and emphasizes the gap between language ability and reliable action (Liu et al., 2024). AgentBoard adds fine-grained progress measures and analytical visualization for multi-turn agents (Ma et al., 2024). SOTOPIA focuses specifically on social intelligence by placing agents in interactive social scenarios and evaluating goal achievement and social dimensions (Zhou et al., 2024). Together, these benchmarks provide controlled tasks, comparative evaluation, and diagnostics for an agent or a small interaction.

GAWorld addresses a complementary unit of analysis: a persistent society run whose agents, environment, institutions, and artifacts evolve across simulated time. Its benchmark layer can check selected outputs, known-sign diagnostics, placebos, and determinism when suitable artifacts exist, but it is not a replacement for standardized social-intelligence or interactive-agent benchmarks. Conversely, task success alone does not validate population composition, emergent dynamics, or policy counterfactuals. GAWorld therefore retains agent-level traces and run-level evidence status so that task competence, social interaction, and society-level claims can be evaluated separately.

3.4 Agent-based social simulation and validation

Agent-based modeling (ABM) treats macro patterns as outcomes of specified micro-level rules and interactions, while generative social science emphasizes using such models to explain how collective regularities can arise (Epstein, 2006). The ODD protocol provides a standard vocabulary for documenting an individual- or agent-based model’s overview, design concepts, and details (Grimm et al., 2006). Methodological work on empirical ABM validation stresses that matching selected aggregate observations is not sufficient: calibration, identification, sensitivity, multiple validation targets, and out-of-sample evidence remain distinct concerns (Windrum et al., 2007; Fagiolo et al., 2007).

GAWorld inherits this demand for explicit mechanisms and layered evidence, but LLM mediation introduces additional sources of variation. Prompt context, provider behavior, model versions, parsing fallbacks, and generated text can change trajectories even when conventional parameters and seeds are recorded. The system therefore couples ABM-style configuration and intervention design to provider logs, raw agent traces, per-arm completion status, and an artifact ledger. Its present archive supports software-path and affordance claims, not empirical



Darker stages are structural boundaries; intermediate stages are configurable and support ablation or insertion.

Figure 1: The configured default agent-step pipeline. The first and last stages are structural boundaries; intermediate stages may be omitted or replaced through configuration. Domain inputs also arrive through direct calls and event-bus contribution points, so the diagram is a control-flow account rather than a claim that every subsystem is a plugin.

validation of a population model. This distinction also responds to concerns that language models should not be treated as drop-in replacements for human participants (Dillion et al., 2023), particularly when generated outputs can flatten variation within identity groups (Wang et al., 2025).

4 Persistent and Situated Agent Lifecycle

GAWorld represents an agent as a continuing participant in a shared simulation, not as an isolated chat session. The representation separates relatively stable profile material, mutable runtime state, and artifacts accumulated over simulated time. Figure 1 shows the connected per-tick control path.

4.1 Profile seeds and mutable state

At construction time, a tabular seed row supplies the identifier, demographic fields, and initial scalar state, while a Markdown profile supplies descriptive material such as occupation, personality, daily life, and values. The current builder merges these sources into a dictionary containing `id`, profile fields, `state`, an in-memory `memory` list, and `social_neighbors`; it then initializes location against the city map. The core state includes affective, socioeconomic, identity, behavioral, and information-exposure variables. Additional mechanisms may add namespaced or compatibility fields during setup and execution.

The class `gaworld.core.agent.Agent` is a typed adapter over this dictionary. It provides properties for identifier, name, state, memory, schedule, location, and attached skill identifiers, while dictionary access delegates to the same backing object. `view_as_agent` therefore introduces no copy, and `ensure_agent_dict` allows a typed caller to hand the record back to older helpers. This is important for incremental migration, but it is not a fully typed agent schema: many runtime call sites still consume dictionaries, and arbitrary mechanism-specific fields can still be added.

Persistence extends beyond the in-memory record. Depending on enabled features, per-agent directories hold episodic entries, diaries, schedules, vector-indexed recall material, environmental preferences, and other mechanism-owned records. On a stateful restart, stored memory is loaded into the agent and can seed retrieval. The file boundary makes inputs to later decisions inspectable, although it also means that reproducibility depends on capturing configuration, external model behavior, and the complete artifact directory rather than only the seed CSV.

4.2 The default per-tick path

The cognition loop is implemented as an ordered `StagePipeline`. A stage receives an agent, a shared per-step dictionary, and the `SimContext`. Hook-visible values retain names such as `activity`, `action`, `outcome`, and `reflection`; intermediate values use underscore-prefixed keys. The default path contains the following twelve stages:

1. **prepare** establishes tick-local inputs and emits the pre-step boundary;
2. **perceive** combines local environment, social, policy, memory, and contributed perception sections;

3. **interrupts** considers events that can alter the scheduled course;
4. **plan** forms the agent’s immediate intention in the current context;
5. **adjust_activity** reconciles that plan with scheduled and interrupted activity;
6. **move** resolves travel and updates location where applicable;
7. **select_action** chooses an action from the available action space;
8. **reflect** produces an interpretation of action and outcome;
9. **update_state** applies state and relationship effects;
10. **broadcast** exposes eligible communication to other agents or a distributed relay;
11. **memorize** constructs and stores the tick’s experience; and
12. **record** appends established logs, state history, and visual trace material.

Before the pipeline runs, the outer loop advances the shared Clock, updates the environment and local occupancy, polls distributed messages when enabled, and may perform configured information seeking. After it returns, the runtime emits `on_agent_post_step`, allowing mechanisms such as the intervention plugin to compute their own metrics. Day-start and day-end events bracket the tick loop, and memory consolidation is invoked at the end of a day. Consequently, the twelve stages are the agent-step spine, not the entire simulation lifecycle.

4.3 Contribution points and compatibility calls

The EventBus lets a mechanism observe an event, contribute zero or more items, or filter a value in priority order. Perception composition is a concrete example: local context can be assembled by established code while plugins add bounded text sections without owning the full prompt. Post-step and day-end events similarly let policy, economy, skills, or interests react at defined boundaries. Observer failures are logged and isolated unless strict mode is enabled; stage failures instead propagate because a missing control-flow stage could silently corrupt state.

Not all inputs use this bus. Memory recall, environment evolution, and social network operations retain direct paths within or around the pipeline. The Controller now validates structured movement requests and exposes audited runtime interventions, but it does not mediate every selected action. Likewise, **record** writes established artifacts in addition to the kernel Recorder. The lifecycle is therefore connected and configurable while retaining compatibility paths beyond the plugin boundary.

4.4 Configuration and ablation semantics

The stage order is read from `CONFIG["pipeline"]["agent_step"]`. Entries may name built-in stages or importable callables; a custom callable can also receive a display name. Unresolvable entries are skipped with a warning, and an empty resolved pipeline is rejected. Downstream built-ins generally read absent step values defensively, which supports mechanism ablations, but **prepare** and **record** are intended as structural boundaries. Changing order or removing a stage can change semantics and artifact completeness. The configuration surface therefore enables controlled system experiments; it does not guarantee that every permutation is meaningful or behavior-preserving.

5 Execution Architecture and Incremental Modularization

GAWorld combines a connected cognition pipeline with a small kernel, nine built-in plugins, and compatibility-preserving domain paths. Figure 2 freezes the audited K1–K5 runtime at repository commit `35a551e`. The archived capability cases predate this migration and are not evaluations of it.

5.1 Kernel services

Simulation context and clock. `SimContext` holds configuration, Clock, EventBus, PluginRegistry, Controller, Recorder, agents and their identifier index, an LLM callable, a seeded random-number generator, and an **extras** compatibility escape hatch. It also provides simulation-level and per-agent plugin-state namespaces. The outer loop is the sole clock writer, giving stages, plugins, and records a common simulated day, time, and tick. This ordering does not by itself make model calls, external services, or concurrency deterministic.

Event bus and plugin registry. The EventBus supports priority-ordered observation (`emit`), merged contribution (`collect`), and value rewriting (`filter`). Handler failures are logged and isolated by default, so warnings and active plugin lists must accompany results. The PluginRegistry assembles built-ins, configuration-named

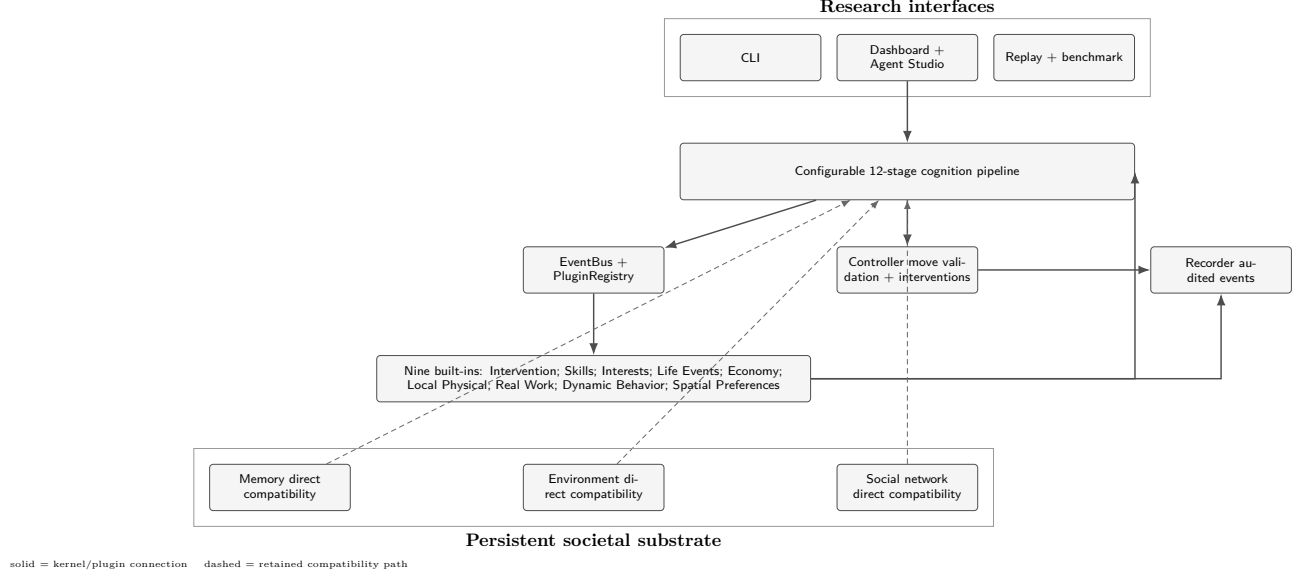


Figure 2: Audited execution architecture at commit 35a551e. Nine built-ins contribute through the registry, bus, pipeline filters, or lifecycle events. Movement uses the Controller validation gate; Recorder events coexist with established domain-specific outputs.

classes, and installed entry points; orders setup by declared dependencies; skips failed or unsatisfied plugins; and tears active plugins down in reverse order. Plugins may own configuration, namespaced agent state, and output directories, but execute in process and are not a sandbox.

Controller. The Controller provides a priority-ordered validator chain over structured `ActionRequests` and a named runtime-intervention API. In the main simulator, movement requests pass through validation: denial preserves the origin and is surfaced in the next perception, while a rewrite can change the destination. K5 registers standard interventions for setting one agent-state value, updating a dotted live-config path, and queueing agent removal for the next day boundary. Denials and interventions are audited through the Recorder. Other selected actions are not all mediated, so the Controller is not a universal policy or safety gate.

Recorder. The Recorder appends clock-stamped JSON objects to table-specific files and is closed when the simulation exits. It records Controller activity and is available to plugins. Main traces, state history, memories, environment events, and specialized metrics also retain established writers. Recorder integration therefore improves cross-plugin auditability without implying that the archive can be replayed from one event log.

5.2 Built-in plugin assembly

The audited registry assembles Intervention, Skills, Interests, Life Events, Economy, Local Physical, Real Work, Dynamic Behavior, and Spatial Preferences. They attach domain behavior to day/tick events, perception contributions, location filters, action or outcome filters, and post-step or day-end hooks. For example, Intervention contributes recommendation context and persists policy metrics, Life Events produces scheduled exogenous events and state effects, Local Physical contributes nearby conditions, and Spatial Preferences can rewrite a resolved destination. The adapters delegate calculations and specialized persistence to established modules: migration changes orchestration and ownership, not the scientific validity of the mechanisms.

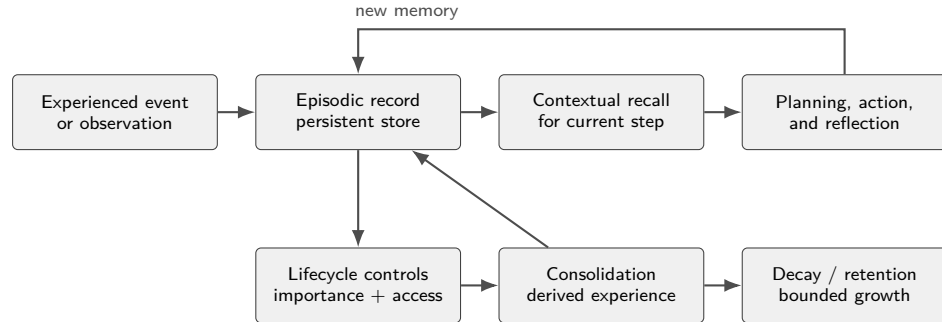
5.3 Migration-state audit

5.4 Failure and reproducibility boundaries

A cognition-stage exception propagates because the stage is control flow. Bus-handler and plugin-setup failures instead warn and normally leave the remaining run active; missing dependencies deactivate the affected plugin.

Surface	Current boundary	Status	Retained boundary and consequence
Cognition	Configured 12-stage pipeline	Wired	Shared step dictionary; order and omissions must be reported
Extensions	EventBus and PluginRegistry	Wired	Failures may degrade rather than terminate a run
Nine domain adapters	Policy, skills, interests, events, economy, physical, work, behavior, spatial	Plugin-integrated	Established stores and writers remain; activation needs config/plugin provenance
Memory	Recall, storage, consolidation, decay	Direct path compatibility	Persistent context is inspectable, not psychologically validated
Environment and social network	World evolution and network modules	Direct path compatibility	The full world and network models are not isolated plugins
Controller	Move validation and named interventions	Connected, scoped	Other actions retain established paths; no universal-gate claim
Recorder	Clocked structured event writer	Connected, non-exclusive	Multiple writers remain; no single-log replay claim

Table 1: Implementation state in the audited repository. K1-K5 completion does not imply that every subsystem or output format is isolated.



Conceptual lifecycle: persistence enables longitudinal state; retention and consolidation prevent an unbounded raw transcript.

Figure 3: Persistent experience and growth paths. Tick-level episodes feed retrieval and rule-based updates; configured day-boundary processes may consolidate episodes into semantic memories or private skills. Solid storage paths denote implemented persistence, not validated models of human remembering or learning.

Operational resilience therefore makes a completed process different from a complete treatment. A reproducible artifact must include resolved stage order, active plugin identifiers, warnings, configuration, source snapshot, model/provider conditions, and specialized outputs.

The scoped K1-K5 migration is complete, but shared dictionaries, the context escape hatch, direct domain calls, and multiple recording paths remain. “K1-K5 complete” describes completion of the migration plan, not universal subsystem isolation or validation.

6 Memory, Cognition, and Behavioral Continuity

GAWorld separates persistence from generation. An agent’s continuity is not held solely in an LLM context window: structured state, episodic records, habits, intentions, relationships, environmental preferences, growth profiles, and skills are written to mechanism-specific stores and reintroduced at later steps. Figure 3 summarizes this path. The resulting system can preserve and expose information across simulated days; whether its memory dynamics resemble those of people is a separate empirical question.

6.1 Episodic persistence and retrieval

The basic store retains an agent’s memory list, schedule, available actions, locations, and location–action biases as JSON, alongside append-only textual logs and simulation state. A second experience layer records episodes in JSONL and persists habits, daily intentions, relationships, and environmental preferences in separate per-agent files. This division supports inspection and selective restart, although a complete restart requires all of these stores rather than a single canonical agent file.

For content-addressed recall, memory entries are also indexed in SQLite with type, simulated time, salience, recall count, and an embedding. The configured embedding path may call an embedding provider; if that path is unavailable, the implementation falls back to a deterministic hash embedding. Retrieval combines vector similarity with salience, age decay, and prior recall, then optionally boosts entries related to the agent’s current growth interests. Thus ranking is a specified scoring procedure over stored entries, whereas the text being ranked may have originated in an LLM-generated action, reflection, diary, news item, or external source.

The day-boundary lifecycle adds two optional transformations. Consolidation selects recent episodic rows and asks an LLM to distill them into a small set of semantic entries; malformed output or insufficient material produces no new entry. Decay is LLM-free: after a configured period it lowers the salience of old, unrecalled episodic rows and deletes rows already below a floor, while protecting semantic, external-information, and procedural entries. A separate episode store also computes exponential salience decay and enforces a size cap. These two decay paths operate over different stores and should not be mistaken for one unified cognitive theory.

External retrieval-augmented memory is likewise optional. Agent-derived queries can retrieve web material, after which an LLM may assess or compress it into external-information entries. This path depends on a caller-supplied fetch adapter and degrades to no operation on failure. Consequently, an experimental report must disclose whether external retrieval, provider-based embeddings, consolidation, and decay were enabled.

6.2 Intentions, habits, and reactive behavior

Behavioral continuity combines model-mediated interpretation with explicit state-transition rules. At day boundaries, recent episodes and current state can be supplied to an LLM to produce priorities, avoidances, social targets, and recovery goals. When the call budget is exhausted or parsing fails, a rule-based fallback constructs intentions from state and recent experience. Habit strength, relationship signals, physiological and time-pressure needs, episode salience, and many mood or stress adjustments are updated by bounded numeric rules. These values enter subsequent prompt context rather than being left as unstructured prose.

The dynamic-behavior module evaluates candidate interruptions from needs, co-location, incoming messages, environmental events, local physical conditions, cascades, and spontaneous urges. Acceptance probabilities depend on event intensity, activity commitment, personality proxies, relationship closeness, and agent state. Candidate generation and schedule insertion are LLM-free but stochastic; they are repeatable only when the relevant random state is seeded. Once an interruption wins, the ordinary activity-adjustment or action path may invoke an LLM to phrase or refine the response. GAWorld therefore distinguishes rule-based opportunity and constraint generation from model-generated interpretation and action text.

Spatial preference illustrates continuity without language generation. A crowd surge or other location anomaly raises a time-bucketed aversion score; scores decay with recency, and a later destination can be redirected when its aversion crosses a configured threshold. This mechanism implements learned avoidance as a transparent rule. It does not establish that the learning rate, threshold, or decay reproduces human spatial choice.

6.3 Interests, skills, and productive activity

Growth profiles contain bounded hobby and skill items with phase, engagement, competence, practice history, and activity templates. Their initial content may be derived by an LLM and cached by a profile signature; deterministic fallback profiles are available when generation fails. Episode matching, practice updates, phase classification, daily retention decay, retirement of stale interests, and socially mediated adoption then use explicit rules and a seedable random generator. The labels borrow a developmental vocabulary, but their implementation is a simulation mechanism rather than a validated model of human interest development.

The skill subsystem maintains a curated global library and per-agent private skills. Available skills can be injected into cognition context, while an optional LLM consolidation step extracts a reusable procedure from recent episodes and saves it as a private Markdown skill. Parsing, schema checks, and registry precedence constrain this output; they do not demonstrate that the procedure is correct or transferable outside the generated setting.

Productive activity extends this pattern to a mock work market. Cached, LLM-derived capability profiles and current growth items inform routing. A JSONL queue provides claim and recovery semantics, a worker pool dispatches briefs concurrently, and code, content, teaching, and web-design adapters use an LLM to produce files. Lightweight validators check properties such as Python syntax, Markdown structure, or HTML/SVG parseability before successful results are ingested into agent state and market settlement. These checks and persisted artifact paths demonstrate a functioning production pipeline; they are not evidence that an artifact meets professional, pedagogical, factual, or security standards.

7 Situated Urban and Societal Substrate

GAWorld places persistent agents in a shared substrate whose components read and write overlapping state. The city constrains movement, local occupancy changes perception, relationships affect encounters, and financial or life events alter later choices. This coupling supports questions that cross individual and aggregate levels, but it also means that mechanism provenance and configuration are part of every experimental treatment.

7.1 City graph and local physical conditions

The city map parses places, roads, transit lines, interiors, and land-use attributes into a graph. Nodes carry categories, coordinates, capacity, opening hours, popularity, density, and display metadata. Destination resolution can match activity terms to place categories, while graph routing and transport-mode rules estimate distance, time, and fare. Mode speeds, fixed delays, road-class multipliers, rush-hour penalties, area price levels, and weather preferences are explicit parameters. They make travel costs inspectable; their presence does not imply transport-model calibration.

At each tick, the local-physical layer recomputes occupancy from stationary agents, clears stale counts, and exposes the current location’s crowding, opening status, and weather. A crowd anomaly is flagged when occupancy is both high and sharply above the previous tick. These snapshots can enter perception and interruption generation, linking collective movement to individual behavior. The mechanism is deterministic given agent locations and parameters, although the locations themselves may follow stochastic or LLM-mediated decisions.

The broader environment system supports configured natural, economic, political, technological, and intraday events. In rule mode it samples from configured event distributions using a local random generator. In LLM mode a model proposes a daily summary, events, and intraday rules that are parsed into the same bounded event schema; configured fallback generation remains available if that call fails. Severity and anomaly tagging are rule-based. Reports must therefore identify the generator mode rather than treating all environmental histories as equivalent.

7.2 Social ties and life events

Relationships are typed records rather than undifferentiated neighbor edges. Role templates specify channels, obligation, protection, and daily decay; interaction signals update closeness and contact time. The implementation can bootstrap off-screen contacts from an LLM-generated backstory, with a deterministic heuristic fallback. It can also generate events involving off-screen contacts, using templates and optional LLM wording, and disclose shared contacts across agents.

Maintenance is rule-based. Uncontacted ties decay at role-specific rates, long-neglected obligations can rise, and an enforcement pass assigns nested Dunbar-style tiers while pruning beyond a configured outer cap and protecting selected kin roles. These tiers are an operational capacity constraint, not evidence that the simulated network reproduces measured human networks.

Life events are normalized, persisted, scheduled by simulated day and time, and drained once when due. Events can target agents and carry bounded state effects and tags. This provides a controlled exogenous-event surface whose timing and direct updates are inspectable. The realism of event frequencies, dependencies, and consequences must be established by a separate study.

7.3 Economic state transitions

The finance module maintains checking, savings, investment, and housing-fund accounts; monthly budgets; income and expense records; liabilities and credit; sector pools; government flows; and macro state. Configured rules calculate social-insurance deductions and progressive income tax, route wages and consumption, enforce available-cash and credit constraints, transfer savings, and settle payments. A dedicated seeded random generator drives

Subsystem	Reads	Writes	Agent-visible consequence	Evidence status
Memory and cognition	Episodes, state, context, retrieved entries	Memories, intentions, habits, relationships	Recalled context, priorities, reflection, and later action bias	Implemented; human realism unvalidated
Interests and skills	Profile, episodes, social growth context	Growth profile, private skills	Practice focus and procedural prompt context	Implemented; partial plugin boundary
World and local physics	City graph, time, weather, locations	Routes, travel state, occupancy snapshots	Feasible destinations, travel costs, crowding and closure cues	Implemented; direct integration
Environment	Configuration, history, optional LLM output	Events, weather and macro context	Interruptions and perceived exogenous conditions	Implemented; generator-mode dependent
Social network	Profiles, contact history, interactions	Typed ties, closeness, obligation, tiers	Encounter probability and relationship context	Implemented; mechanism validity unassessed
Life events	Persisted event queue, clock	Consumption flag and configured state effects	Scheduled exogenous changes to targeted agents	Implemented; committed plugin-integrated event producer
Economy	Activity, balances, calendar, sectors, macro state	Accounts, ledger, budgets, pools, agent state	Income, spending constraints, taxes, credit, returns and payments	Implemented; primarily hook-based
Work	Capability cache, activity, growth, market jobs	Queue results, artifacts, settlement and state effects	Produced files, earnings, and experience records	Implemented; artifact quality unvalidated

Table 2: Coupling among agent and societal components. Evidence status describes implementation visibility, not empirical validity.

expenditure variation, market returns, shocks, and other probabilistic transitions so that unrelated random draws need not perturb the economy sequence.

Shared sector and government pools make counterparties visible instead of creating every monetary flow without an account. Agent-to-agent payments and some rent or labor flows can be routed to modeled recipients; investment returns combine a common market component with idiosyncratic variation. Ledger and conservation checks can reveal implementation defects such as unbalanced internal transfers. They cannot establish that the tax schedule, consumption response, labor process, credit behavior, or macro dynamics is an empirically valid representation of a real economy. Accounting consistency is therefore a software invariant, not a behavioral-validation result.

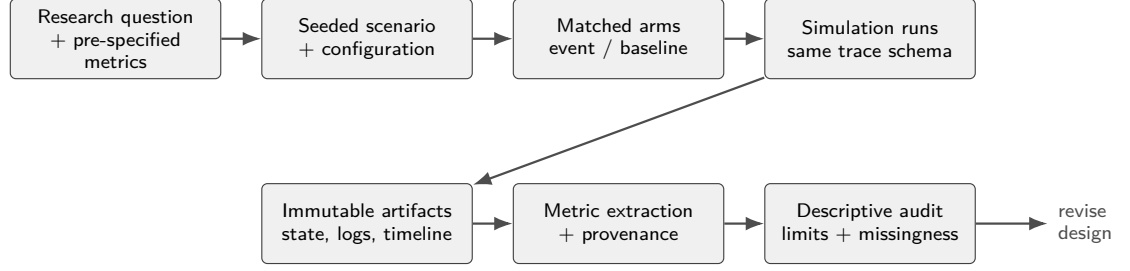
7.4 Coupling and ownership boundaries

Table 2 makes the principal shared-state paths explicit. “Implemented” refers to an inspectable current mechanism; “partial boundary” indicates that the mechanism is connected but retains direct calls, shared dictionaries, or specialized writers rather than a fully isolated plugin interface.

The table also exposes the current modularity limit. Interests and skills have plugin entry points, and economy, local physical perception, spatial preferences, work, and dynamic behavior now have built-in adapters. Memory, environment evolution, and social-network operations still exchange mutable agent dictionaries or runtime objects directly. This arrangement is usable and inspectable, yet replacing one mechanism can change prompts, state schemas, random draws, and specialized artifacts elsewhere. Controlled comparisons should therefore freeze the complete coupled configuration rather than naming only the focal intervention.

8 Experimental Workflow and Evidence Production

GAWorld treats an experiment as a configured execution that produces an inspectable artifact tree, rather than as a prompt followed by a single model response. The ordinary entry point, `generative_city_sim.py run`, loads the selected agents and city map, restores persistent state when enabled, assembles the currently integrated plugins, and advances a shared simulation clock. Runtime configuration controls the horizon, agent cohort, random seed,



Evidence boundary: a workflow trace is not a real-world causal estimate.

Figure 4: GAWorld’s experiment path separates a research question and configured scenario from matched execution arms, stored traces, metric extraction, and a descriptive audit. The final boundary is substantive: an inspectable comparison is not by itself a real-world causal estimate.

provider routing, environment, policy events, and output locations. A run can therefore be reconstructed at the level of its declared configuration and stored traces, although exact replay also depends on external model behavior, network services, and the completeness of the archived inputs.

8.1 Matched Event and Baseline Runs

For intervention studies, the `compare-event` command constructs two scenario directories under a timestamped comparison root. The `without_event` arm inherits the configured policy-event list, whereas the `with_event` arm appends a named event at an explicit simulation day and clock time. Both arms receive the same requested agent subset, simulation horizon, random seed, and—when supplied—LLM provider override. Before execution, each arm is reset within its own isolated memory, log, vector-database, state, network, environment, intervention, and visualization directories. The two runs are then launched concurrently. This organization reduces accidental cross-arm state leakage and makes missing or failed arms visible through separate reset and run logs.

After both subprocesses finish, the comparison routine reads each arm’s `state/agent_state_history.csv`. For every available metric, it computes the across-agent final snapshot, the mean over recorded observations, and event-minus-baseline differences. Machine-readable results are stored in `comparison_metrics.csv`; `comparison_summary.md` provides a human-readable view. The separate `run_meta.json` records choices such as seed, horizon, provider, and whether the lower-fidelity fast mode was requested. Policy-oriented fields can include stance, toxicity, misinformation risk, cross-viewpoint exposure, and intervention reward when their upstream mechanisms emit them. Importantly, archived policy runs contain fields that remain identically zero, including misinformation-risk and cross-viewpoint-exposure series in some comparisons. Presence in the schema must therefore not be reported as evidence that the corresponding mechanism was active.

8.2 What Matching Does—and Does Not—Identify

Sharing a pseudorandom seed and configuration is a software control, not a causal-identification strategy. LLM endpoints can be nondeterministic despite fixed sampling parameters; concurrent calls may experience different failures, fallbacks, latency, or service-side model revisions; and an injected event can change subsequent prompts and thereby amplify trajectory divergence. Moreover, the implemented report contrasts raw trajectories without random assignment, replication, uncertainty estimation, or adjustment to an external population. Consequently, GAWorld’s paired output supports debugging, mechanism tracing, and hypothesis generation. It does not establish that a measured difference was caused by the analogous intervention in a real city, or even that a single-run simulation contrast is stable across seeds.

A defensible study should therefore pre-specify outcomes and event timing, archive the exact configuration and provider information, repeat both arms over multiple seeds, include negative controls where appropriate, and report missingness and failures alongside effect summaries. The event should also be traced to the subsystem expected to implement it: textual exposure to a policy is not equivalent to a verified economic, mobility, or network-state

update. This distinction is especially important in an LLM-based society, where a plausible narrative response can mask a disconnected structural mechanism.

8.3 Heterogeneous but Inspectable Outputs

The current runtime writes several complementary views of a simulation. Agent logs preserve local action and cognition traces; state histories provide a long table of agent, step, metric, and value; environment output contains the shared timeline and contextual events; memory directories contain durable episodic, relationship, growth, and vector-retrieval state; and the economy subsystem maintains financial snapshots and audit artifacts. Visualization traces encode frames for replay, while intervention plugins retain policy-specific metrics. Comparison roots place two complete scenario trees beside their derived summary files. Together these artifacts allow an analyst to move from an aggregate difference back to the relevant agents, times, and subsystem records.

This plurality is currently both a strength and a limitation. The emerging kernel provides an append-only **Recorder** that time-stamps plugin events into named JSONL streams, but it has not replaced or normalized the simulator’s existing logs, CSV files, environment timeline, memories, and subsystem-owned outputs. Cross-subsystem analysis still requires explicit joins and attention to distinct schemas and clocks. We therefore describe GAWorld as inspectable, not as having a fully unified provenance store.

8.4 GAWorld-Bench as an Audit Layer

GAWorld-Bench consumes existing output trees or, when explicitly requested, orchestrates comparison runs. Its implemented tracks examine macro-level anchors and selected intervention diagnostics, including known-sign checks, placebo handling, and an optional determinism comparison. The scorecard marks unassessed components rather than treating missing evidence as success, and labels its composite as a diagnostic summary rather than a scientific result. Other proposed validation dimensions remain unimplemented or incomplete. Synthetic fixtures bundled with the harness exercise parsers and scoring paths; they are software tests and are never treated in this paper as observations from an artificial society. The benchmark thus complements the artifact workflow in [figure 4](#): it helps expose coverage gaps and suspicious outputs, but cannot repair an underpowered or poorly identified experiment.

9 Research Interfaces and Deployment Modes

GAWorld exposes the same persistent society through several interfaces, each optimized for a different stage of research. These interfaces are intentionally thin over repository state and runtime commands: they improve inspection and operation, but do not constitute independent simulation engines.

9.1 Command Line and Interactive Inspection

The command-line interface is the most direct research surface. In addition to **run**, **reset**, and **compare-event**, it can restrict the simulation horizon, import external retrieval material, create an agent from provided social-media text, serve visualization artifacts, and start local services. The **interview** command rebuilds a selected agent from the configured population, restores its memory when statefulness is enabled, seeds retrieval from that memory, and answers one or more researcher-supplied questions. Interviews are therefore probes of a stored agent state under a new elicitation context; they are not independent measurements of the human profile on which an agent may have been based.

LLM access is mediated by a provider router. Current adapters cover local Ollama, OpenAI-compatible, and Anthropic-style endpoints, while routing rules can select providers by task or agent and can specify ordered fallbacks. Calls are logged with provider, task, agent, prompt size, latency, and fallback position. This abstraction makes deployment adaptable to local or hosted models, but it does not make providers behaviorally interchangeable. A model or routing change is an experimental change and should be recorded as such.

9.2 Dashboard and Agent Studio

The local Dashboard serves a static web client together with HTTP endpoints for configuration, agent lists, profiles, state, memory, life events, simulation start/stop status, trace metadata, and interviews. It supports configuring provider routing and initiating a run while exposing the resulting files to a researcher. Agent Studio extends this

surface with an agent-centric view: it can inspect and edit profiles and normalized state variables, show memory and growth summaries, inspect skills and relationships, and launch an interview. These tools make persistent state legible without requiring an analyst to read every JSON or CSV file directly.

The web interface should not be conflated with the kernel’s **Controller**. The latter validates structured movement and exposes named, audited runtime interventions. The Dashboard’s primary run and mutation paths still use HTTP handlers and CLI subprocess management rather than routing all operations through that API.

9.3 Trace Replay and External Environment Service

The simulator’s visualizer records a simulation trace and latest frame, including agent positions, activities, events, and presentation assets. `site/simviz` replays these artifacts in a browser and provides spatial and temporal inspection distinct from aggregate plots. Replay is valuable for finding implausible schedules, stalled movement, or mismatches between an event and an agent response. It remains a view of what was recorded: omitted state cannot be recovered from the animation, and visual plausibility is not a validation criterion.

An optional external-environment server separates shared world evolution from the main process. Its HTTP API starts simulation days, advances time ticks, returns environmental events and context, and persists engine state plus day/tick caches. It can use the configured LLM for environment generation or a rule-based fallback. This service boundary supports controlled reuse of a world backend and makes environment state independently inspectable; the local in-process environment remains a supported runtime path.

9.4 Relay-Based Distributed Communication

GAWorld also implements an opt-in distributed communication mode. Simulation nodes register local agents with an HTTP relay, discover agents in the same cluster, poll bounded inbound messages, and send bounded outbound messages. The relay persists its directory and retained message state, while the simulation converts received messages into context for local cognition. The CLI can host the relay directly, and the Agent Studio can expose detected relay activity.

This mechanism is best understood as a communication bridge, not as evidence of scalable distributed simulation. The current implementation does not provide distributed clock synchronization, partitioned world-state ownership, transactional consistency, fault-tolerant consensus, or published throughput and scaling measurements. It enables agents hosted by different processes or machines to exchange messages through a shared service; claims about speedup, population scale, or consistency under failure require separate evaluation.

9.5 Deployment Boundaries

The default services use Python threaded HTTP servers and file-backed state, which keeps local research deployment transparent and easy to inspect. Hosted LLM credentials are supplied through environment variables rather than embedded in profiles or manuscript artifacts. Researchers exposing Dashboard, environment, or relay services beyond a trusted machine must add production authentication, transport security, access control, rate limiting, and data governance appropriate to the profiles and generated content in use. GAWorld’s current interfaces are research tooling; their existence should not be read as a production-security or large-scale-operations claim.

10 Existing Capability Cases

This section audits what the archived outputs demonstrate about GAWorld as a research instrument. It does not treat an available file as a successful scientific evaluation. We use *complete* only when an artifact is adequate for the narrow descriptive purpose stated here; *partial* when arms, durations, metrics, or interactions are missing; *designed* when a protocol exists without usable outcome evidence; and *fixture* for synthetic data that tests analysis code. These labels describe evidence availability, not system performance.

10.1 Case 1: Persistence Is Observable, but the Memory Contrast Is Not

The memory-consistency archive is useful because it exposes a longitudinal experimental shape that is often hidden in agent demonstrations. One agent (ID 34) was assigned to four intended treatments: intact memory, reset between phases, selective summary deletion, and injected conflict. Each arm planned a seven-day acquisition phase followed by a seven-day continuation phase. The archive contains seven phase-1 diaries for every arm, together

Capability area	Archive status	What the current artifacts support
Macro economy + wellbeing	COMPLETE	Multivariate longitudinal trace production
Persistent-memory study	PARTIAL	Cross-phase artifacts; incomplete treatment coverage
Policy comparisons	PARTIAL	Matched event/baseline workflow; single-run evidence
Information interventions	PARTIAL	Exposure and stance metric pipeline; inactive fields disclosed
Network evolution	PARTIAL	Initialization trace; interactions insufficient
Emotion contagion	DESIGNED	Protocol exists; archived outcome evidence incomplete
ABM validation	DESIGNED	Validation workflow exists; completed evidence insufficient
GAWorld-Bench scorecard	FIXTURE	Software-path diagnostics only, not simulation findings

Statutes describe evidence availability, not comparative performance or scientific validity.

Figure 5: Coverage of the audited result archive. The matrix distinguishes inspectable capability evidence from incomplete evaluations and diagnostic fixtures; it is not a leaderboard or a validation score.

with state and memory files. It also records different memory-file footprints, including episodic, growth, habit, and relationship artifacts. Thus a researcher can inspect what persisted, what was deleted, and how a later elicited action was represented.

Only the intact-memory arm, however, contains all seven phase-2 diaries. The reset, selective, and conflict arms each stop after phase 1, with the archived report attributing the interruption to initialization after the worker pool started. The archive therefore demonstrates file-backed persistence across two phases for one agent in one arm; it does not compare the behavioral effect of memory retention, deletion, or conflict. Qualitative labels such as a move from “procrastinating” to “active” are post-hoc codings of a few generated actions, not a validated continuity measure. In particular, differences in phase-1 final actions cannot explain missing phase-2 outcomes. This failure is scientifically informative at the infrastructure level: treatment completeness must be checked before interpreting a fluent diary as longitudinal evidence.

10.2 Case 2: Policy Artifacts Exercise a Counterfactual Workflow

The policy archive contains four timestamped comparisons—traffic restriction, housing subsidy, medical reimbursement, and job-training subsidy. Each stores an event arm beside a baseline arm and exports the same metric columns, thereby exercising the isolation and reduction path described in [section 8](#). All are single archived comparisons, with events injected on simulation day 3 (09:00 for traffic and 08:00 for the other three); there is no multi-seed sampling distribution or uncertainty estimate.

The raw CSV files also show why mechanistic inspection must precede substantive interpretation. In every pair, the final values of `cross_viewpoint_exposure`, `misinformation_risk`, `stance_score`, and `toxicity_score` are zero in both arms. Some generic state variables differ: for example, the housing comparison has a final event-minus-baseline `time_pressure` difference of -0.0311 and the traffic comparison a `mobility_intent` difference of -0.0036 . These numbers are reported only to show that the reducer retains metric direction and magnitude. A single LLM-mediated trajectory cannot tell whether the named policy caused the difference, whether the relevant structural mechanism was connected, or whether the sign would persist under a new seed. The case supports a workflow claim—GAWorld can preserve paired trees and compute declared contrasts—rather than a claim about housing, health, training, or transport policy.

10.3 Case 3: Information Metrics Make Inactivity Visible

The misinformation artifact contains three five-agent arms. The control and treatment B each cover two simulated days, whereas treatment A covers one. Across every recorded day and arm, average misinformation risk is 0.0, and

Table 3: Status of all experiment families represented in the frozen artifact ledger. Counts and horizons describe the archived files, not a common evaluation protocol.

Area	Status	Archived scope	Permitted use
Macro economy and well-being	Complete for description	$n = 5$ agents, one seed, three-day configuration; 116 recorded steps per agent	Multivariate state-trace production, not an economic or psychological finding
Persistent memory	Partial	$n = 1$ agent; four planned 14-day arms; only one completed both seven-day phases	Persistence and cross-phase artifact inspection, not a treatment comparison
Policy comparisons	Partial	Four single archived event–baseline pairs; Day-3 events; no seed replications	Pairing, isolation, and metric extraction, not policy effects
Information intervention	Partial	Three arms, $n = 5$ each; unequal horizons of one or two days	Information-metric schema and missing/inactive-field audit
Polarization	Partial	Two arms with 109 versus 85 records	Availability of a comparison artifact; unequal records preclude an effect claim
Network evolution	Partial	$n = 5$, 14-day report; zero recorded interactions on every day	Initialization and failure visibility, not emergent-network evidence
Emotion contagion	Designed / incomplete	Four intended arms; all four report a missing state-history file	Protocol coverage only; no outcome result
ABM validation	Designed / incomplete	Validation workflow is present; completed empirical evidence is insufficient	Evaluation design only, not an empirical validation result
GAWorld-Bench	Fixture	Built-in synthetic scorecard	Parser and scoring-path diagnostics only

the reported peak and final risks are also 0.0. Mean cross-viewpoint-exposure is 0.06656 for control, 0.06707 for treatment A, and 0.06635 for treatment B. The unequal horizons rule out a balanced comparison, and differences at the fourth decimal place are descriptive properties of these files, not estimated intervention effects.

That negative result is nevertheless a useful system diagnostic. The archive records treatment names, horizons, agent counts, daily risk series, and an exposure aggregate together. It does not prove that misinformation propagated, that risk detection was sensitive, or that the interventions changed exposure. The separate polarization artifact reinforces the boundary: its two arms contain 109 and 85 records, respectively, and thus cannot be read as a balanced longitudinal contrast without reconstructing the missingness. GAWorld’s value here is inspectability—inactive fields and unequal coverage remain visible rather than being silently converted into a success score.

10.4 Case 4: Economy–Wellbeing Traces Provide Dense State, Not External Validity

The macro-economy archive is the most complete artifact for demonstrating multivariate trace production. Its hashed long-format CSV contains 11,600 rows for five agents, 116 unique recorded steps per agent, and 20 metrics per step ($5 \times 116 \times 20$). The associated configuration describes a three-day, seed-42 run. Direct aggregation of the CSV gives 580 values for each metric across agents and steps. Mean emotion, stress, and economic security are 0.4896, 0.5490, and 0.6270, respectively; their observed ranges are [0.0233, 0.9310], [0.0907, 0.9898], and [0.4455, 0.8298].

The raw row count matters because the accompanying narrative report describes “580 timesteps per agent.” In the CSV, 580 is instead the number of agent–step values for one metric (5×116); each agent has 116 distinct step identifiers. We therefore use the CSV cardinalities rather than the report’s timestep wording. This discrepancy illustrates the role of an artifact ledger: even a plausible derived report must be checked against its hashed source.

The trace can support new descriptive analyses without a new simulation. For example, partitioning the 116 step identifiers into consecutive ranges 0–38, 39–76, and 77–115 yields emotion means of 0.6365, 0.5020, and 0.3305, while stress means are 0.4122, 0.5428, and 0.6916. These bins are equal-index summaries, not validated “days,” and the trajectories remain generated state variables from one seed. They show that GAWorld records joinable, time-indexed state across agents and constructs; they do not show human wellbeing dynamics, economic realism, or an effect of any real-world shock.

Table 4: Claim–evidence matrix for the audited GAWorld snapshot. “Current support” states the strongest defensible claim, not an aspirational goal.

Claim level	Evidence required	Current support	Not established
Individual consistency	Repeated independent elicitations; longitudinal and perturbation tests; external or held-out criteria	Persistent profiles, memories, diaries, interviews, and one completed two-phase arm are inspectable	Human fidelity, stable identity across models/seeds, or a memory-treatment effect
Macro correspondence	Pre-specified external targets; calibrated population; uncertainty and out-of-sample tests	State/economy pipelines can produce aggregates; benchmark fixtures test analysis code	Population representativeness, calibrated macro fit, forecasting, or real-city prediction
Emergent dynamics	Multiple seeds and scales; mechanism ablations; network/temporal diagnostics; robust qualitative patterns	Social and network mechanisms emit inspectable state; failures are archived	Polarization, contagion, homophily, or other stable emergent laws
Counterfactual response	Replicated randomized arms; verified intervention path; placebo and negative controls; uncertainty	Four paired event/baseline trees demonstrate comparison mechanics	Causal policy effects, sign stability, transportability, or welfare recommendations
Reproducibility	Immutable code/config/data/model snapshots; deterministic components; repeated reruns and failure accounting	Hashed result files, seeds/configs in some runs, file-backed traces, and a diagnostic harness	Bitwise replay of hosted LLM calls or complete end-to-end reproduction of every archive

10.5 Failures Are Part of the Capability Record

Two additional archives prevent the coverage matrix from becoming a showcase of successful-looking outputs. The network-evolution report describes five nodes and two initialization edges over a nominal 14-day horizon, but records zero agent–agent interactions on each of the 14 days; the edges were inferred from initial co-location. It cannot support a homophily or emergent-network claim. The emotion-contagion comparison contains four arm entries, all of which report that their state-history CSV is missing. It contains no contagion outcome. Finally, the GAWorld-Bench scorecard matches its built-in synthetic fixture. It verifies that parsers, checks, and report generation execute; it is not a result from GAWorld agents. Reporting these cases alongside the more complete trace is essential to the scientific use of an inspectable system.

11 Validation Boundaries

An artificial society is not validated by the realism of a transcript, by one aggregate match, or by successful execution. GAWorld separates five claim levels: individual consistency, macro correspondence, emergent dynamics, counterfactual response, and reproducibility. Evidence at one level does not compensate for absence at another. This layered view follows established agent-based-modeling practice in distinguishing implementation verification, empirical validation, and the intended domain of use (Grimm et al., 2006; Windrum et al., 2007; Fagiolo et al., 2007).

11.1 Individual Consistency

GAWorld implements the prerequisites for a consistency study: named persistent agents, structured state, episodic files, recall, reflection, and longitudinal logs. These mechanisms allow within-agent questions such as whether a stated preference survives intervening events or whether an interview answer is traceable to stored experience. The current memory case does not answer those questions comparatively because three of four arms lack phase 2 and the only complete arm contains one agent. Generated behavioral coherence is also not equivalent to correspondence with a real person. A stronger evaluation would repeat standardized probes before and after controlled memory perturbations, separate temporal persistence from prompt sensitivity, and compare against held-out observations where consent and provenance permit.

11.2 Macro Correspondence

Aggregating simulated variables is an implemented capability; calling the aggregate realistic requires a separate calibration argument. The audited economy trace has five agents and one seed, while the benchmark scorecard is a synthetic diagnostic fixture. Neither represents a target population or supports out-of-sample prediction. Future macro validation should document the sampling frame, map each simulated construct to an external measurement, reserve targets for testing rather than calibration, and report uncertainty across seeds and plausible parameterizations. Agreement on a few marginal means would remain insufficient if joint distributions, temporal patterns, or subgroup errors failed.

11.3 Emergence and Mechanism

An emergent claim requires a pattern that is not merely written into a prompt or initial condition and that recurs under controlled variation. The present archive does not meet that standard: the network study records no interactions, the emotion study has no state files, and the polarization arms have unequal record counts. The existence of social-network, exposure, and affect fields therefore establishes instrumentation, not a social law. Appropriate tests would vary population size and topology, ablate candidate mechanisms, quantify between-seed variability, and trace a macro pattern back to micro-level events. Where a field remains identically zero, the first question is whether the mechanism and detector were active, not why society produced no effect.

11.4 Counterfactual Validity

The matched-run interface is a useful scaffold for counterfactual experiments, but the archived policy differences are single contrasts produced by an LLM-mediated dynamical system. A common seed does not guarantee common latent randomness after prompts diverge, and a policy description does not guarantee that an economic or mobility mechanism changed. Before interpreting a difference, researchers should verify the intervention’s path from event to structural state, replicate both arms over seeds, pre-register target metrics, and include placebo events or negative-control outcomes (Lipsitch et al., 2010). External policy conclusions additionally require a defensible mapping to the population and institutions of interest. GAWorld currently supports constructing and auditing such a study, not replacing it.

11.5 Reproducibility as a Non-Compensatory Gate

The artifact hashes used in this paper bind each numerical statement to an audited byte sequence. They do not bind the archive to a clean released runtime: the audit began from a dirty working tree, and hosted providers may change behavior independently of code, seed, or sampling parameters. Missing state files and interrupted arms further show that successful completion must be recorded per treatment. Consequently, a result that cannot be reconstructed from code, configuration, inputs, model identity, and complete outputs should not be promoted because it looks plausible or matches an external statistic.

For future releases, the minimum reproducibility unit should contain the exact code revision plus a patch or clean tag, environment lockfile, initial profiles and maps, provider/model identifiers, prompt and sampling settings, random seeds, per-arm status, logs, and raw and derived artifacts. Hosted-model studies should estimate replay variability rather than promise bitwise determinism. The diagnostic benchmark can then serve as a regression gate: fixtures test parsers and scoring logic, while separately labeled simulation runs test whether scientific evidence has actually been produced.

11.6 Intended Use

At its present evidence level, GAWorld is best used for mechanism prototyping, trace inspection, hypothesis generation, controlled software experiments, and the design of more rigorous artificial-society studies. The archive does not support deploying GAWorld as a substitute population, forecasting real social outcomes, ranking policies, or making decisions about identifiable people. Those are not merely stronger wordings of the same result; they are different claims requiring external data, identification, uncertainty quantification, and governance that the current artifacts do not provide.

12 Limitations, Ethics, and Roadmap

GAWorld makes a broad set of mechanisms available in one runtime, but breadth does not imply scientific validity or engineering completeness. We separate limitations of the current artifact archive, limitations of the system implementation, and limitations inherent in representing social life through LLM agents. The resulting roadmap likewise separates engineering work that improves the instrument from research work needed to justify substantive claims.

12.1 Evidence and reproducibility limitations

The archived experiments are incomplete and heterogeneous. Several contain a single seed or one agent, treatment horizons differ, some mechanisms produced identically zero fields, and some intended arms lack outcome files entirely. The evidence ledger can make those conditions visible, but it cannot recover missing treatments or provide statistical power after the fact. The existing cases therefore illustrate trace production, comparison mechanics, and failure visibility rather than stable behavioral effects. Claims about individual consistency, macro correspondence, emergence, or counterfactual response require purpose-built evaluation beyond the archive summarized here.

Reproducibility is also limited by model and provider dependence. A seed does not freeze a hosted model, its serving stack, safety filters, or routing behavior. Provider fallback may change the model used within a run; library, prompt, or parser versions can change subsequent state; and concurrency can alter call order. Local models improve control but do not eliminate hardware, runtime, and numerical variation. Consequently, meaningful replication requires more than source code and a seed: it requires resolved configuration, prompt and sampling settings, provider and model identifiers, dependency versions, active plugins, warnings, per-arm completion status, and raw artifacts. Hosted-model studies should measure replay variability rather than promise bitwise determinism.

Runtime and monetary cost constrain the studies that can be performed. A persistent society may invoke generation, embedding, retrieval, reflection, environment, and consolidation paths repeatedly across agents and ticks. Model caching, rule-based fallbacks, configurable stages, and local providers can reduce calls, but each changes experimental conditions. Cost pressure can also encourage small populations and short horizons that are inadequate for emergence or uncertainty analysis. Future evaluations should report call counts, latency, failures, model routes, and estimated cost alongside simulated scale and horizon.

12.2 Architecture and data limitations

The K1–K5 microkernel migration is complete for its defined scope, including nine built-in plugins, movement validation, and audited standard interventions. Compatibility paths nevertheless remain: memory, environment, and social logic are not wholly plugin-owned, the Controller does not mediate every action, and the Recorder is not the sole provenance stream. Shared dictionaries and the context escape hatch weaken schema enforcement and ownership boundaries. Third-party plugins run in process and should not be treated as sandboxed. A configured extension point therefore does not guarantee mechanism isolation, security, or semantic equivalence under replacement.

The artifact layer has corresponding schema plurality. Agent logs, long-form state CSVs, environment timelines, memory stores, visualization traces, economy audits, and plugin outputs use different schemas and sometimes different notions of step or time. This plurality preserves domain-specific detail, but cross-subsystem analysis requires explicit joins and can silently misread cardinality or missingness. The discrepancy identified in the economy case between report wording and raw CSV cardinality illustrates this risk. Until a versioned provenance schema and manifest bind clocks, entities, mechanisms, and derivations, replay and analysis must remain artifact-aware.

Input profiles are another boundary. They may be sparse, stylized, synthetic, or assembled from data whose completeness and provenance vary. Rich generated backstories do not repair missing observations; they can turn uncertainty into plausible detail. Profile coverage, source, consent, transformation, and missingness should be documented, and downstream readers should distinguish supplied attributes from model-generated elaboration.

12.3 Ethical boundaries and affected communities

Synthetic agents can create an illusion of access to perspectives that were never actually observed. Language models may reproduce stereotypes, suppress within-group heterogeneity, and make marginalized identities appear more uniform or legible than they are (Wang et al., 2025). Conditioning a model on demographic labels or a short profile does not confer lived experience, community membership, consent, or authority. Even when aggregate

outputs resemble survey responses in one setting, LLM agents should not be treated as drop-in replacements for human participants (Dillion et al., 2023).

GAWorld therefore must not substitute for engagement with people affected by a research question or policy. It cannot determine community priorities, adjudicate contested values, estimate acceptable harms, or legitimate a decision about housing, health, employment, policing, mobility, or information governance. For consequential applications, affected communities should help define the question, categories, outcomes, unacceptable uses, and release conditions; domain experts and empirical data are needed to evaluate whether the modeled mechanisms correspond to the institutions at issue. Simulation may support deliberation or identify assumptions to test, but it cannot replace participation, due process, or accountable public decision-making.

Profiles, memories, interviews, and generated social ties can also expose or infer sensitive information. Researchers should minimize collection, obtain appropriate consent and permissions, separate identifiers from research artifacts, restrict access, define retention and deletion policies, and assess whether releasing prompts or traces could re-identify a source. Dashboard, relay, and environment services are local research tooling rather than production-secure deployments; exposing them requires authentication, encryption, authorization, audit, and abuse controls. Generated claims about real people should not be presented as observations, and simulated agents should not be used to rank, target, or make decisions about identifiable individuals.

12.4 Engineering roadmap

The engineering roadmap is to make current boundaries enforceable and current artifacts easier to reconstruct. Priorities are: (1) extend typed Controller validation beyond movement where governance is required; (2) migrate remaining domain mechanisms incrementally behind declared dependencies without changing their semantics silently; (3) replace shared dictionary conventions with versioned schemas and explicit state ownership; (4) unify trace identifiers and clock semantics through a manifest-backed provenance layer while retaining domain-specific payloads; (5) capture model, provider, prompt, fallback, dependency, and plugin resolution in every run; and (6) add conformance, failure-injection, migration, security, and replay tests. These changes would improve composability and auditability. They would not, by themselves, validate an artificial society.

12.5 Research roadmap

The research roadmap begins only after the execution unit is reproducible. For individual consistency, standardized repeated probes and controlled memory perturbations should be evaluated against held-out criteria where consent and provenance permit. For macro correspondence, researchers should define a sampling frame, map simulated constructs to external measures, separate calibration from testing, and quantify subgroup error and uncertainty. Claims of emergence require multiple seeds, scales, topologies, and mechanism ablations, with macro patterns traced to micro events. Counterfactual studies require verified intervention paths, replicated arms, pre-specified outcomes, placebos or negative controls, and an explicit account of transportability. Provider and prompt sensitivity should be treated as experimental factors, not hidden implementation noise.

Finally, future studies should evaluate not only whether a simulated pattern is reproducible, but also whether the representation and intended use are appropriate. Participatory problem formulation, impact assessment, and documentation of excluded perspectives belong beside technical validation. GAWorld’s near-term role is therefore to make artificial-society hypotheses, mechanisms, traces, and failures easier to construct and inspect—not to turn generated populations into unearned evidence about society.

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